l-Diversity for Top-Down Greedy Anonymization

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1 Introduction

k-anonymity [3] guarantees that every record shares its quasi-identifier (QI) values with at least k-1 other records, so a person cannot be linked to a single row. It does not say anything about the sensitive attribute (SA), though. If all records of an equivalence class (EC) have the same SA value, an attacker who knows the QIs of a person learns the sensitive value anyway (homogeneity attack). l-diversity [1] closes this gap by requiring that every EC contains at least l "well-represented" SA values.

For this exercise we took the Top-Down Greedy (TDG) algorithm from the kaylode k-anonymity toolbox [4] and changed it so that it produces tables that are both k-anonymous and l-diverse. We then compared the modified algorithm with the original TDG on the Adult dataset. Already at k = 10 the original TDG leaves 4 ECs where all people have the same occupation and 45 ECs with fewer than 3 occupations; at k = 2 there are 1,046 homogeneous ECs covering 2,404 people.

2 Method

2.1 Top-Down Greedy in the toolbox

TDG [2] starts with the whole table as one partition and splits it recursively. To split a partition it picks two records u and v that are far apart. The distance is the NCP of the generalisation that covers both records, and instead of checking all pairs the toolbox uses a heuristic with three rounds (start at a random record, jump to the farthest one, repeat). Every other record goes to the seed it is closer to. If one side ends up with fewer than k records, the algorithm either moves the nearest records from the other side or merges both sides again, whichever has the lower NCP. Partitions with fewer than 2k records are not split any more, and in the end each partition is generalised to the lowest common ancestor in the hierarchies.

2.2 Our modification

We added the parameters l, the sensitive attribute and the type of l-diversity. A split is only accepted if both halves are k-anonymous and l-diverse. We implemented the three definitions from [1]:

- **distinct l-diversity:** at least l different SA values in the EC,
- **entropy l-diversity:** the entropy of the SA distribution in the EC is at least log(l),
- **recursive (c, l)-diversity:** $r_1 < c (r_l + r_{l+1} + ... + r_m)$, where r_i is the i-th most frequent SA count.

The simplest version would be to reject a split as soon as one half is not l-diverse. Because TDG splits by QI distance only, this happens very often and the partitions stay large (see Section 4.3). So instead we try to repair the split, similar to the k-balancing that TDG already does:

1. Take the half that is not valid (the other one is valid, otherwise the split is rejected).
2. Sort the records of the valid half by their NCP distance to the invalid half.
3. Go through them and move a record over only if the invalid half still needs records to reach k or the record improves its diversity (new SA value, higher entropy or larger recursive margin), and the valid half is still k-anonymous and l-diverse without it.
4. Stop as soon as the invalid half is valid. If this does not work, or the repaired split has a higher total NCP than not splitting, the partition is kept as one EC.

Since we check at the start that the whole table is l-diverse and only valid partitions are ever created, every EC in the output fulfils k and l. With l = 0 the original k-balancing is used, so the plain topdown method behaves like before. Besides that we fixed the random seed of the pair heuristic (so runs are reproducible) and a global list that was not reset between runs. The new method is called topdown_ldiv and is started with, e.g., python anonymize.py --method=topdown_ldiv --k=10 --l=3 --ldiv=entropy --dataset=adult. Unit tests check on the Italia dataset that every EC has at least k records, is l-diverse and that no SA value was changed.

3 Experimental setup

We used the Adult dataset of the toolbox (30,162 records). QIs are sex, age, race, marital-status, education, native-country and workclass with the hierarchies from the toolbox. The SA is occupation with 14 values; the most frequent one (Prof-specialty) covers only 13.4% of the records, the rarest (Armed-Forces) 9 records. In the toolbox occupation was a QI, we removed it from the QIs for all methods so that the comparison is fair. We ran three experiments:

- **A:** k = 10, l from 2 to 8 (distinct), 2 to 6 (entropy) and 2 to 4 (recursive, c = 2), compared with TDG at k = 10.
- **B:** l = 3 (distinct), k in {2, 3, 5, 10, 20, 50, 100} for both methods.
- **C:** k = 10, l = 3 and 6, split repair switched off (split is just rejected).

Information loss is measured with NCP, the discernibility metric (DM), the normalised average EC size (CAVG) and the number of ECs. For privacy we count homogeneous ECs (only one SA value) and ECs that violate 3-diversity. For classification we used the toolbox pipeline (based on [5]): KNN, SVM and Random Forest are trained on the anonymised training split (80%) to predict salary-class and evaluated with weighted F1 on the anonymised test split. Generalised values are replaced by the mean of the range or a one-hot encoding of all leaves. We had to fix some things in this evaluation: non-generalised values were dropped from the features, the F1 score ignored classes that were never predicted, the label order was random and the SVM had no feature scaling. All numbers below use the fixed version.

4 Results

4.1 Privacy

Table 1: Experiment B, original TDG vs. TDG with distinct 3-diversity for different k (Adult, SA = occupation).

k	TDG ECs	homog. ECs (records)	ECs < 3 values	TDG NCP	TDG+l ECs	violations	TDG+l NCP
2	6,587	1,046 (2,404)	4,522	0.018	3,866	0	0.045
3	4,730	298 (1,056)	1,564	0.032	3,800	0	0.043
5	3,145	70 (409)	350	0.052	2,978	0	0.055
10	1,784	4 (49)	45	0.091	1,758	0	0.092
20	1,041	0 (0)	1	0.142	1,041	0	0.142
50	441	0 (0)	0	0.251	441	0	0.251
100	234	0 (0)	0	0.373	234	0	0.373

Table 1 shows that the original TDG is clearly vulnerable for small k: at k = 2, 4,522 of 6,587 ECs have fewer than 3 occupations. The number goes down with larger k, but even at k = 20 there is still 1 EC with only 2 values. Only at k = 50 is the output 3-diverse "by accident", and then the NCP is 0.251. Our method reaches 3-diversity at k = 10 with an NCP of 0.092, so 2.7 times less information loss than increasing k. topdown_ldiv had zero violations in every run, for all three l-diversity types.

The effect is easy to see on the small Italia dataset (SA = disease, k = 2, l = 3, see experiments/italia_demo_k2_l3.md): the original TDG creates 5 ECs where both people have the same disease, e.g. age 40~50 from Veneto with heart disease, and 35 of its 41 ECs have fewer than 3 diseases. With topdown_ldiv there are 20 ECs and all of them contain at least 3 different diseases.

4.2 Information loss

For small l the extra cost is almost zero (Figure 1): with distinct l = 3 the NCP goes from 0.091 to 0.092 and the number of ECs from 1,784 to 1,758. This is because occupation is quite evenly distributed, so most ECs already contain several values and only the few bad splits need repairing. From l = 6 on the cost grows faster (NCP 0.112 at l = 6, 0.163 at l = 8). Entropy l-diversity is stricter than distinct l-diversity with the same l, because a single dominant occupation is not allowed: entropy l = 5 (0.168) costs about as much as distinct l = 8 (0.163), and entropy l = 6 already gives NCP 0.268 with 678 ECs. Recursive diversity with c = 2, l = 3 costs 0.116, a bit more than entropy l = 3. CAVG follows the NCP, DM is noisier but shows the same trend. At k = 10 the original TDG took 27 s and topdown_ldiv between 19 and 41 s, at k = 2 it was 52 s vs. 69 s. These are single runs on a

laptop and a second run gave quite different times (the anonymised tables themselves were identical), so we can only say that there is no large overhead. This is expected, since the repair only needs one extra pass over a partition when a split fails.

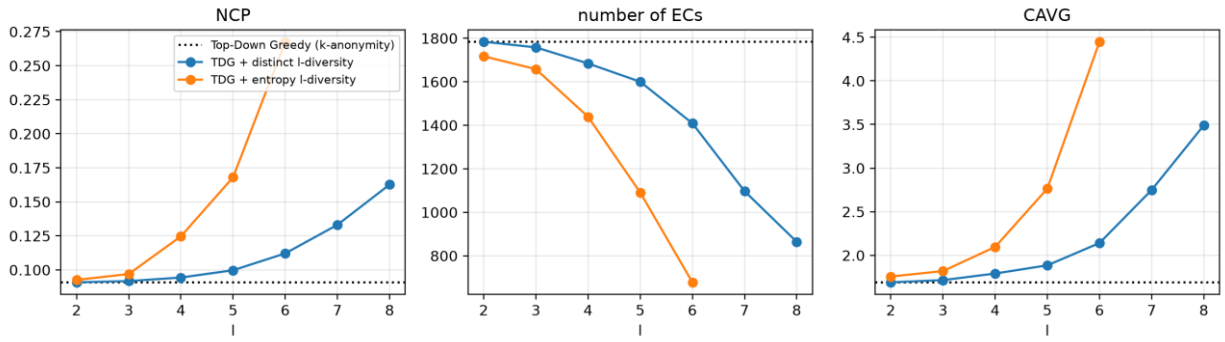


Figure 1: Experiment A (k = 10). NCP, number of ECs and CAVG for increasing l. The dotted line is the original TDG at k = 10.

4.3 Why the repair step matters

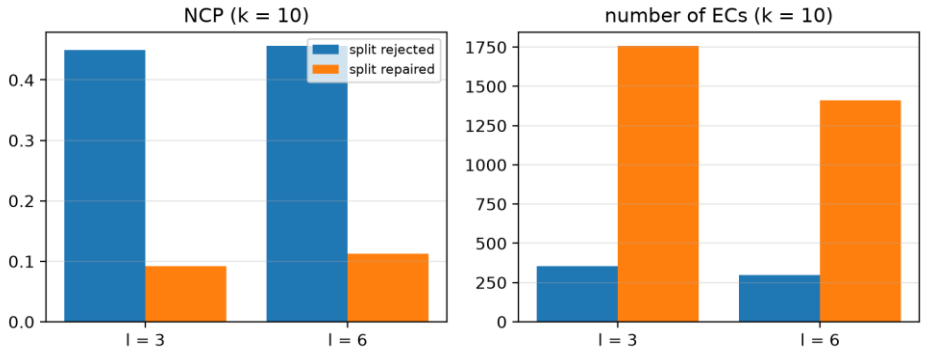


Figure 2: Experiment C (k = 10). Rejecting invalid splits vs. repairing them.

If invalid splits are just rejected, the NCP is 0.450 for l = 3 and 0.456 for l = 6, with only 354 and 298 ECs (Figure 2). With repair it is 0.092 and 0.112. The reason is that TDG decides splits only by QI distance, so already near the root one half is often missing some occupations. Rejecting the split there stops the recursion for a very large part of the table. Moving a handful of records with the missing SA values is enough to keep splitting.

4.4 Classification

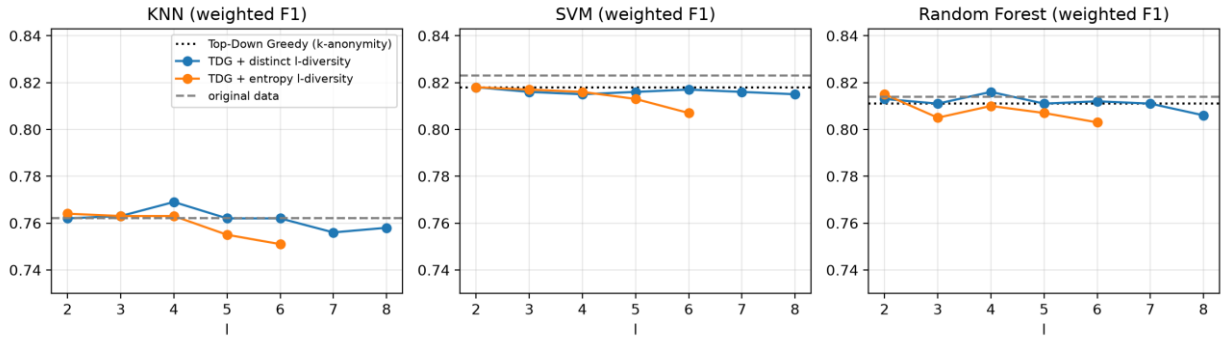


Figure 3: Experiment A (k = 10). Weighted F1 on salary-class; grey = original data, dotted = original TDG.

On the original data the F1 scores are 0.762 / 0.823 / 0.814 (KNN / SVM / RF). The original TDG at k = 10 reaches 0.762 / 0.818 / 0.811, and with distinct l = 3 we get 0.763 / 0.816 / 0.811, which is the same within noise. Even in the strongest setting (entropy l = 6) the scores only drop to 0.751 / 0.807 / 0.803. Over all our runs topdown_ldiv was never more than 0.021 below the original data. Occupation itself is not used as a feature, and the few extra generalisations that the repair causes are spread over many ECs, so the classifiers barely notice them. In experiment B the largest F1 difference between the two methods at the same k was 0.013.

5 Discussion and conclusion

Adding l-diversity to TDG was mostly a question of where to put the check. Checking only at the end of a split is easy but wastes a lot of utility; repairing the split with records that bring in missing SA values keeps TDG's good information loss. On Adult, 3-diversity on top of 10-anonymity is practically free (NCP +0.001, same F1), while the original TDG needs k = 50 to get there. Higher l and entropy l-diversity become expensive quickly, so l should be chosen together with the SA distribution.

Limitations: we only used one dataset and one SA, and occupation is relatively balanced, so on a skewed SA the cost would be higher. The repair is greedy and does not look for the best set of records. Distinct l-diversity does not protect against skewness or similarity attacks (e.g. three different but similar jobs); for this, t-closeness would be the next step. Finally, the whole table has to be l-diverse at the start, otherwise the algorithm stops with an error.

References

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[2] J. Xu, W. Wang, J. Pei, X. Wang, B. Shi, A. Fu. Utility-Based Anonymization Using Local Recoding. KDD 2006.

[3] L. Sweeney. k-Anonymity: A Model for Protecting Privacy. IJUFKS 10(5), 2002.

[4] M.-K. Pham. kaylode/k-anonymity toolbox, https://github.com/kaylode/k-anonymity (TDG from https://github.com/fhstp/k-AnonML).

[5] D. Slijepcevic et al. k-Anonymity in Practice: How Generalisation and Suppression Affect Machine Learning Classifiers. Computers & Security, 2021.